

Influence of Chinese herbal plants on greenhouse gas emissions from pig manure composting and fermentation

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ARTICLE INFO

Keywords:

Chinese herbal plant

Radix isatidis

Radix Polygoni Multiflori

Pig manure composting

Mitigating greenhouse gas emissions

ABSTRACT

To reduce greenhouse gas emissions during the composting of animal manure and promote the resource utilization of Chinese herbal plants, experiments on pig manure composting were conducted in a plastic greenhouse. A mixture of herbal powders (*Radix isatidis* and *Radix Polygoni Multiflori*, mixed in a 1:1 ratio) was added to the pig manure at concentrations of 0.05%, 0.1%, and 0.5% (w/w), respectively. The results showed that the emission peaks of greenhouse gases occurred in the early stages of composting. The addition of 0.5% of a mixture of Chinese herbal plant powders resulted in a significant reduction in cumulative emissions of CO₂ (45.0%) and N₂O (60.0%) compared to the control. This treatment also resulted in the lowest global warming potential, measured at 91.3 g kg⁻¹, and significantly increased the germination index. The study concluded that adding 0.5% Chinese herbal plant powder to compost effectively mitigated greenhouse gas emissions during the composting process.

1. Introduction

Since the implementation of reform and opening up in China, the economy has developed rapidly, leading to a significant rise in people's incomes and living standards. Consequently, the structure of people's diets has changed considerably. The increased demand for meat and dairy products among Chinese residents has led to the large-scale and intensive development of the livestock and poultry farming industry. As of 2022, the number of poultry, pigs, and cattle slaughtered reached 16.1 billion, 700.0 million, and 48.4 million, respectively (National Bureau of Statistics of China, 2022), generating approximately 3.8 billion tons of livestock and poultry manure annually (Yang et al., 2023). Livestock and poultry manure contains significant amounts of organic matter, nitrogen, phosphorus, potassium, and trace elements necessary for plant growth. Composting such manure may make fertilizer, improve the physical and chemical properties of the soil, increase soil organic matter, sequester carbon, and contribute to mitigation efforts (Fan et al., 2023; Kraut-Cohen et al., 2023; Li et al., 2023; van den Bergh et al.,

2023), Producing organic fertilizer from livestock and poultry manure is considered beneficial not only to the country but also aligns with current Chinese policies to "ensure food security," "ecological security," and "comprehensively promote the revitalization of the countryside".

A large amount of greenhouse gases (GHG), such as CH₄, CO₂, and N₂O, is emitted during the composting of livestock and poultry manure (Cáceres et al., 2018; Bai et al., 2020; Chen et al., 2022; Wang et al., 2024). It has been reported that total carbon and nitrogen losses during the composting process reached 48.7% and 27.5%, respectively. The total carbon loss due to volatilization in the form of CO₂ and CH₄ was 34.8% and 0.9%, respectively, while the total nitrogen loss due to volatilization in the form of N₂O was 1.5% (Zhang et al., 2021). The emissions of GHGs led to the loss of nutrients in organic fertilizers (Awasthi et al., 2020a), decreased fertilizer efficiency (Huang et al., 2022), and increased the global warming potential (GWP) (Cao et al., 2019; Liu et al., 2021). Although the emissions of CH₄ and N₂O during the composting process were small, their 100-year warming potential will be 28 and 265 times that of CO₂, respectively (Shakoor et al., 2021;

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<https://doi.org/10.1016/j.ibiod.2024.105882>

Received 5 June 2024; Received in revised form 8 August 2024; Accepted 11 August 2024

Available online 17 August 2024

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Y. Liu et al., 2023), resulting in a significant warming potential. This issue has been receiving increasing attention. Ren et al. (2023) found that the greenhouse effect due to the emissions of CH₄ and N₂O during the composting process of pig manure ranged from 102.0 to 127.3 g kg⁻¹. The significant amount of GHG emissions underscores the importance of decreasing GHG emissions in the composting process.

GHG emissions are affected by compost temperature, pH, moisture content, C/N ratio, and turning frequency (Harindintwali et al., 2021). Composting is a dynamic process of chemical reaction, and GHG emissions from composting are often influenced by a combination of factors (Y. Liu et al., 2023). CH₄ is produced by methanogenic bacteria under anaerobic conditions. Methane production increases when the compost has high moisture, low porosity, and vigorous microbial metabolism, all of which lead to insufficient oxygen content inside the heap (Ge et al., 2018; Harrison et al., 2022; Lei et al., 2022).

The optimization of key composting process parameters, such as oxygen supply, C/N ratio, and humidity, is helpful for the rapid decomposition of compost and the reduction of GHG emissions (Hoang et al., 2022; Tang et al., 2023). Applying additives to compost materials may also be considered an effective measure for further reducing GHG emissions (Godlewska et al., 2017; Ba et al., 2020; Nguyen et al., 2022; Yu et al., 2023). This includes physical, chemical, and biological additives, with physical additives seeming to provide better emission reduction for physical additives (Hoang et al., 2022). Physical additives can reduce GHG emissions from composting by improving compost porosity and capacity, adsorbing or directly participate in carbon and nitrogen conversion, influencing key microbial communities and environmental factors, and enhancing the activity of metabolism-related enzymes thereby aiding in C and N conversion (Agyarko-Mintah et al., 2017; Awasthi et al., 2020b; Hoang et al., 2022; Nguyen et al., 2022). Biochar has attracted considerable attention because of its large specific surface area, rich pore structure, strong adsorption capacity, and remarkable emission reduction effects (Liu et al., 2020; Nguyen et al., 2022; Yin et al., 2024). However, large-scale applications are limited due to the high cost of preparation and other constraints.

China is the largest producer of traditional Chinese medicine, generating a large amount of traditional Chinese herbal waste every year, which urgently needs to be reused (Zhu et al., 2022). Bardon et al. (2014) found for the first time that proanthocyanidins (flavonoids) can inhibit GHG emissions in extracts of *Polygonum multiflorum* Thunb (*Fallopia* spp.). This group also confirmed in a previous study that *Radix isatidis*, which contains flavonoids, is sufficient to inhibit GHG emissions from farmland soil (Wu et al., 2017). However, there has been no reports on the addition of Chinese herbal plants to reduce GHG emissions from livestock and poultry manure fermented to produce organic fertilizers. Hence, it is very meaningful and necessary to investigate whether and how Chinese herbal plants can decrease GHG emissions from composting animal manure.

In the present investigation, a hypothesis was proposed that Chinese herbal plant powders would inhibit GHG emissions from animal manure composting. The goal is to investigate whether and how the Chinese herbal plant powders of *Radix Polygoni Multiflora* (*Polygonum multiflorum* Thunb) and *Radix isatidis* mitigate GHG emissions from composting pig manure.

2. Materials and methods

2.1. Raw material collection and characteristics

In this study, pig manure and rice straw were used as raw materials for composting. The pig manure and rice straw were purchased from Huai'an, Jiangsu Province. The Chinese herbal plants, *Radix isatidis* and *Radix Polygoni Multiflora*, were purchased from China (Bozhou) Traditional Chinese Medicine Trading Center. The composition of the compost raw materials is shown in Table 1.

Table 1

Characteristics of composting materials used in this work.

	Moisture (%)	TOC (g·kg ⁻¹)	TN (g·kg ⁻¹)	C/N
Pig manure	55.26	301.13	21.64	13.91
Rice straw	6.43	374.25	10.17	36.78
<i>Radix isatidis</i>	8.49	455.30	3.99	114.10
<i>Radix Polygoni Multiflora</i>	7.07	416.46	5.76	72.35

2.2. Experiment design

The composting experiment was conducted at the Agricultural Experiment Station of Nanjing University of Information Science and Technology, China, with four treatments and three replications for each treatment. Both Chinese herbal plants (*Radix isatidis* and *Radix Polygoni Multiflora*) were added to pig manure to co-ferment, aiming to test whether and how the Chinese herbal powders impact GHG emissions during pig manure composting. Rice straw was utilized to adjust the compost C/N ratio to about 25 with a moisture of 60%. The experimental design included four treatments of Chinese herbal powder: (1) 0% (CK), (2) 0.05% (BH1), (3) 0.1% (BH2), and (4) 0.5% (BH3) of the dry weight of the compost. The dried roots of both Chinese herbal plant were ground and passed through a 1.2 mm sieve, then blended at a ratio of 1:1. Pig manure and rice straw were initially mixed well, then blended with the Chinese herbs several times and placed in a 60 L foam box (inner dimension L × W × H of 49.2 cm × 49.2 cm × 32.2 cm) for composting and fermentation. Each mixture weighed 22 kg. All the boxes were placed inside a plastic greenhouse.

2.3. Pile temperature detecting, compost and GHG sampling

2.3.1. Compost sampling

The pig manure mixture was composted for a total of 30 days. The manure pile temperature and ambient temperature were measured at 9:00 a.m. and 4:00 p.m., respectively, using a mercury thermometer. The pig manure pile in the box was turned and re-mixed at 3, 7, 15, and 30 days after the start of the composting. Samples were collected on days 0, 3, 7, 15, 30.

2.3.2. GHG sampling

GHGs were collected on days 1, 3, 5, 7, 11, 15, 20, and 30 after composting using the static box method. The round cylinder was made from transparent plastic with a size of 20 cm (inside diameter) × 29 cm (height). A small hole (1 cm in diameter) was drilled in the top center of the cylinder, and a rubber plug was inserted into the hole. The cylinder was then inserted into the center of the compost pile to a depth of 5 cm to seal it. Sampling began immediately after the cylinder was sealed. While sampling gases, a syringe was inserted through the rubber plug at the top of the cylinder, and the syringe was pumped up and down 10 times to fully mix the gas in the static cylinder. Sampling was performed every 15 min, with a total of three collections made. The thermometer was hung inside the cylinder to monitor the air temperature. The collected gases were stored in a 20 mL gas collection bag.

2.4. GHG determination

The GHG (CO₂, CH₄, and N₂O) samples inside the plastic bags were determined using a gas chromatograph (7890B, Agilent Technologies Co., Ltd., America) equipped with an electron capture detector (ECD) and a hydrogen flame ionization detector (FID). Pure N₂ (99.999%) was used as the carrier gas and pure N₂ with 10% CO₂ served as the buffer gas for the ECD. The temperature of the ECD was set to 300 °C, and the column oven temperature was set to 60 °C. Pure CO₂, CH₄, and N₂O gases (purchased from Nanjing Sanyuan Industry Gas Company Ltd.,

Nanjing China) were used as standard gases to create and calibrate the GHG standard curve for reference.

2.5. Moisture, pH, GI (germinating index), ammonia and nitrate determinations

The wet compost was weighed and air-dried to a constant weight. Both dry and wet weights were measured to determine the compost pile moisture. Wet compost samples were mixed with distilled water at a ratio of 1:10 (w/v) and shaken at 30 °C for 1 h (Yang et al., 2022). The liquid was then filtered to collect the filtrate. The pH of the filtrate was determined using a pH meter. Five milliliters (5 mL) of the filtrate were pipetted into a 9 cm Petri dish lined with filter paper. Subsequently, 10 full and uniform radish seeds (uncoated) were placed on the filter paper and incubated at 30 °C for 48h under light protection. After the cultivation was completed, the germination rate and root length were measured. Using distilled water as a control, the germination index was calculated according to equation (1) (Luo et al., 2018). The compost was extracted with 2 mol L⁻¹ of KCl to determine inorganic nitrogen (NH₄⁺-N and NO₃⁻-N), and measured using a flow analyzer (Seal AutoAnalyzer 3, Germany) (Tang et al., 2023).

$$GI = \frac{A_1 - A_2}{B_1 - B_2} \quad (1)$$

Where A₁ and B₁ represent seed germination (%), A₂ and B₂ represent root length (mm) of treatment and control, respectively.

2.6. GHG calculation

The gas emission rate based on the change in gas concentration in the box over time was calculated, equation (2):

$$F = \frac{273}{273 + T} \times \frac{M}{V_m} \times \frac{dC}{dT} \times \frac{V}{m} \quad (2)$$

where F denotes the gas emission rate (flux), mg•kg•d⁻¹; T is the incubation temperature (°C); M is the molar mass of CO₂, CH₄ or N₂O (g•mol⁻¹); V_m is the molar volume of the gas (CO₂, CH₄ or N₂O) in the standard state (L•mol⁻¹); $\frac{dC}{dT}$ is the change in gas concentration inside the chamber over time; V is the headspace volume of the static chamber (m³); m is the dry weight of composting material (kg).

Cumulative GHG emissions were estimated using linear interpolation, equation (3):

$$Q = \sum_{i=1}^n \frac{F_i + F_{i+1}}{2} \times t_i \quad (3)$$

Where Q is the cumulative gas emission during the 30-day composting process (mg•kg⁻¹); n is the number of gas sampling times; F_i and F_{i+1}: the compost gas emission rates for the ith and i+1st sampling times, respectively (mg•kg•d⁻¹); t_i: the number of days between two consecutive sampling times.

The global warming potential (GWP) is the equivalent amount of GHG converted using CO₂ as the reference gas, based on the warming potential of CH₄ and N₂O relative to CO₂ on a 100-year scale (1 for CO₂, 28 for CH₄, and 265 for N₂O).

$$GWP = Q(\text{CO}_2) + 265 \times Q(\text{N}_2\text{O}) + 24 \times Q(\text{CH}_4) \quad (4)$$

Where Q(CO₂), Q(N₂O), and Q(CH₄) represent the cumulative emissions of CO₂, N₂O, and CH₄, respectively, in the composting process.

2.7. Data process and statistical analysis

The experimental data were analyzed and graphed using Excel 2021. One-way analysis of variance (ANOVA) was performed to test for significant differences between treatments using R (version 4.3.3) with a

95% confidential level. The results are expressed as the mean ± SD of three replicates.

3. Results and discussion

3.1. Effects of Chinese herbal plant powder on pig manure compost pile temperature, moisture, pH and germination index

3.1.1. Temperature

During the composting process, microorganisms are continuously involved in the conversion of organic matter and heat release, which affects the composting temperature (Chen et al., 2018). Temperature can be used to visually assess microbial activity in the compost (Czekala et al., 2016). In this study, the temperature changes in all compost treatments showed similar patterns (Fig. 1A). The compost temperature began to increase after the start of composting. In treatments BH2 and BH3, the temperature peaked on day 4, reaching 36.8 °C and 45.0 °C, respectively. In treatment BH1, the highest temperature of 42.3 °C was recorded on day 5, which was attributed to the highest microbial activity and rapid decomposition of the organic matter at this stage. (Niu et al., 2024). However, in treatment CK, the maximum temperature of 40.3 °C was reached on day 10. The results indicate that the addition of herbs improved the porosity of the compost, thereby stimulating the fermentation of pig manure and causing the pile temperature to rise 5–6 days earlier (Zhou et al., 2018).

3.1.2. Moisture

The overall decreasing trend in moisture during the composting process is shown (Fig. 1B), which is consistent with the results of Agyarko-Mintah et al.(2017). At the start of composting (day 0), the moisture content ranged from 59.2% to 61.4%, depending on the treatment, which was conducive to the smooth progress of composting (Zhang et al., 2021; Tang et al., 2023). At the end of composting, the lowest moisture content of 56.0% was observed in treatment BH1, which is significantly lower than that in CK (58.5%) ($p = 0.018$). In treatment BH1, the moisture content was reduced by 8.7% compared with that on day 0 ($p < 0.01$). This reduction is attributed to increased evaporation as the temperature of the fermentation pile rose.

3.1.3. pH

The pH of the compost initially increased, then decreased, and finally stabilized (Fig. 1C). Similar findings were reported by Chen et al. (2020), where the initial increase in pH may be attributed to the rapid conversion of NH₄⁺-N and the decomposition of organic acids. In the later stages of composting, the decrease in pH could be due to nitrification processes. The initial pH of all treatments ranged from 6.9 to 7.3, which is considered to be on the lower end of the pH range for GHG emissions (Zhang et al., 2021). After the composting started, the pH of the pig manure pile increased, similar to the results of previous studies (Qiu et al., 2021; Yang et al., 2022). The pH peaked at day 3 reaching 7.2 to 7.4. From days 3–7 of composting, the pH decreased, and on day 7, the lowest pH of 6.80 was observed in the treatment BH2. At the end of composting, the pH of the treatments ranged from 6.98 to 7.2, with no significant difference between treatments. This indicates that the addition of Chinese herbal powder did not impact the pH of the pile.

3.1.4. Germination index (GI)

GI is an index to assess the maturity of animal manure composting. A trend of initially decreasing and then increasing GI was observed across different treatments (Fig. 1D), which was similar to the trend reported in the study by Chen et al. (2018). In the early stages of composting, extensive mineralization and decomposition of organic matter led to an increase in soluble salt content, a decrease in GI, and an enhancement of toxic effects on seeds (Awasthi et al., 2020b; Yang et al., 2023). As composting proceeded, the GI value increased continuously, reaching 81.5 %–115.0 % on the 30th day. The germination index of BH3 was

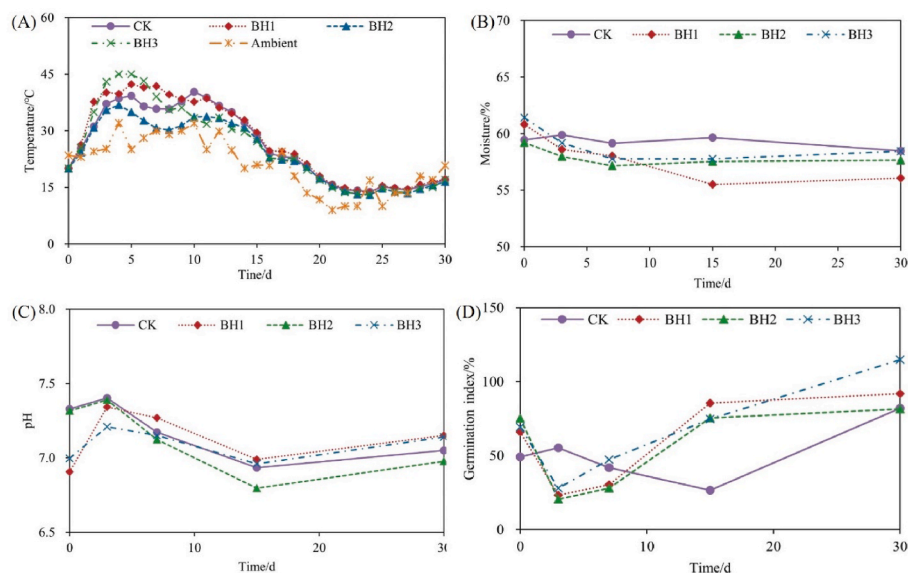


Fig. 1. Variations of (A) pile temperature, (B) pile moisture, (C) pile pH and (D) germination index of pile during the composting process.

significantly higher than that of CK ($p = 0.047$). When $GI > 80\%$, there was no toxic effect on plants (Yang et al., 2022). Therefore, the addition of 0.5% herbal powder improved the quality of compost.

3.2. Effects of Chinese herbal plant powder on ammonia and nitrate of pig manure composting

The trend of $\text{NH}_4^+\text{-N}$ content across the four treatments is shown in Fig. 2A. This trend is opposite to that reported by Liu et al. (2024), but consistent with the results of Yang et al. (2023). $\text{NH}_4^+\text{-N}$ content decreased rapidly in the early stages of composting (except in treatment CK), probably because nitrification converted $\text{NH}_4^+\text{-N}$ to $\text{NO}_3^-\text{-N}$. In treatment CK, the $\text{NH}_4^+\text{-N}$ content increased from $2010.7 \text{ mg kg}^{-1}$ to $2575.0 \text{ mg kg}^{-1}$ between days 0 and 3 of composting, which was significantly higher than in the other three treatments ($p < 0.001$). It was possibly ascribed to the lack of herbal powder in treatment CK, resulting in poor aeration condition inside the pile, and the inhibition of nitrification, which led to the accumulation of $\text{NH}_4^+\text{-N}$ (Kuypers et al., 2018; Wang et al., 2022, 2024). Subsequently, the trend of $\text{NH}_4^+\text{-N}$ content changes in each treatment became similar and stabilized. At the end of composting, the $\text{NH}_4^+\text{-N}$ contents across the four treatments ranged from 37.3 to 97.8 mg kg^{-1} , with the three treatments containing herb powder having significantly higher levels than CK ($p < 0.05$).

As shown in Fig. 2B, the $\text{NO}_3^-\text{-N}$ content showed an overall increasing

trend during the composting process, which was consistent with the results of previous studies (Jiang et al., 2018; Yang et al., 2023). The $\text{NO}_3^-\text{-N}$ content in treatment BH2 was generally higher throughout the composting process, probably because the temperature in this treatment was lower, which was suitable for the growth of nitrifying bacteria. As a result, $\text{NH}_4^+\text{-N}$ was converted to $\text{NO}_3^-\text{-N}$ more rapidly (Jiang et al., 2018). Finally, the $\text{NO}_3^-\text{-N}$ content of treatment BH1 was significantly lower than that of the other three treatments ($p = 0.030, 0.026, 0.036$). Thus, the addition of 0.5% of herbs may promote denitrification during the composting process.

3.3. Effects of Chinese herbal plant powder on GHG emissions from pig manure composting

3.3.1. CO_2 emissions

Aerobic composting is a process in which microorganisms metabolically decompose organic matter in the substrate into stable small molecules (CO_2 , H_2O) and convert them into humus (Czekala et al., 2016; Q. Liu et al., 2023). Studies have shown that CO_2 is inevitably produced during the composting process (Czekala et al., 2016; Harindintwali et al., 2021). A higher rate of CO_2 emission indicates a faster degradation of organic matter in the compost (Chen et al., 2018). The CO_2 emission rate and cumulative emissions under different additions of Chinese herbal powder are shown in Fig. 3. CO_2 emissions were concentrated in

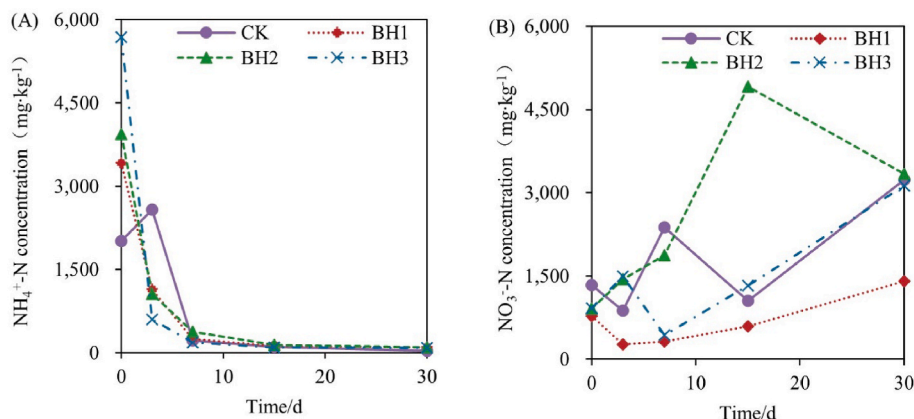


Fig. 2. Variations of (A) pile $\text{NH}_4^+\text{-N}$, (B) pile $\text{NO}_3^-\text{-N}$ during the composting process.

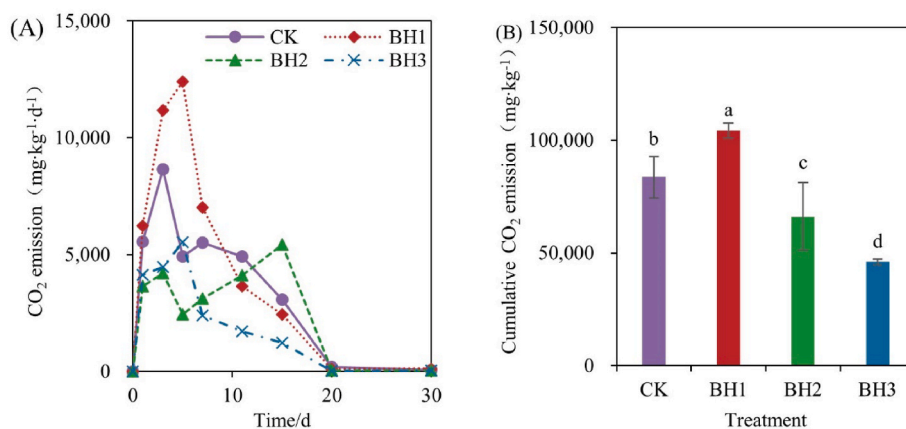


Fig. 3. The emission rate and cumulative emission of CO₂ during pig manure composting.

the early stages of composting (Fig. 3A), which was consistent with the results of other studies (Chen et al., 2018; Ottani et al., 2023). The highest CO₂ emission rates for treatments CK, BH1, BH2, and BH3 were 8640.1 mg kg⁻¹ d⁻¹ (3rd day), 12398.1 mg kg⁻¹ d⁻¹ (5th day), 5422.8 mg kg⁻¹ d⁻¹ (15th day), and 5522.0 mg kg⁻¹ d⁻¹ (5th day), respectively. The peak in CO₂ emissions observed during the early stages of composting was mainly due to the increase in composting temperature, vigorous microbial metabolism, enhanced enzyme activity, and rapid degradation of organic matter (Czekala et al., 2016; Wang et al., 2022). The temperature change curve also supports this viewpoint (Fig. 1A).

By analyzing the cumulative CO₂ emissions of the four treatments, it was found that the cumulative CO₂ emission in treatment BH1 was significantly higher than that of CK (83609.6 mg kg⁻¹) ($p = 0.022$). The cumulative CO₂ emissions in treatments BH2 and BH3 were significantly lower than that of CK ($p < 0.05$), with reductions of 20.9% and 45.0%, respectively. This indicates that the addition of 0.05% of herbal powder to the compost promoted CO₂ emissions, while the addition of 0.1% and 0.5% of herbal powder inhibited CO₂ emissions.

3.3.2. CH₄ emissions

Methane is mainly produced by methanogenic bacteria under anaerobic conditions (Ge et al., 2018; Awasthi et al., 2020a). The CH₄ emission rates and cumulative emissions with different additions of Chinese herbal powder are shown in Fig. 4. Large quantities of CH₄ were emitted during the first 15 days of composting, which may be due to the high activity of aerobic microorganisms consuming a large amount of oxygen and creating anaerobic areas within the compost pile. This led to increased activity of methanogenic bacteria and the production of large quantities of methane (Ge et al., 2018; Yu et al., 2023; Ottani et al.,

2023). However, a low CH₄ emission rate was observed throughout the composting cycle, which may be attributed to certain active biochemical factors in Chinese herbal powder suppressing methanogen activity.

The cumulative CH₄ emissions from the four treatments ranged from 0.5 to 7.1 mg kg⁻¹, which are consistent with the cumulative emission levels reported in other investigations (Lei et al., 2022). The cumulative CH₄ emissions from treatments BH1 and BH3 were significantly higher than those of CK, while the cumulative CH₄ emissions in treatment BH2 were significantly lower than those of CK ($p < 0.001$), with a reduction of 80.2%. Thus when the addition amount was 0.1%, the compost may have had a better pore structure, lower methanogenic bacteria activity and higher methane-oxidizing bacteria activity, resulting in lower CH₄ emissions.

3.3.3. N₂O emissions

Organic nitrogen is converted to nitrate through ammonification and nitrification under oxygen-sufficient conditions, and part of the nitrate is denitrified to produce N₂O (Cáceres et al., 2018; Kuypers et al., 2018). The N₂O emission rate and cumulative emissions are shown in Fig. 5. The N₂O emission rate peaked at either 1–3 days, except for treatment BH2, where the N₂O emission rate was highest on day 3 (59.8 mg kg⁻¹ d⁻¹). In contrast, the remaining three treatments had their highest N₂O emission rates on day 1, ranging from 44.4 mg kg⁻¹ d⁻¹ to 111.5 mg kg⁻¹ d⁻¹. Among them, the peak N₂O emission rates in treatments CK and BH1 were significantly higher than those in treatments BH2 and BH3 ($p < 0.05$). This may be due to the decrease in NO₃⁻-N content in treatments BH2 and BH3 day 3 of composting (Wang et al., 2024). The N₂O emission rate then gradually decreased and reached its lowest at day 20. Afterward, the emission rate of N₂O increased again, similar to the

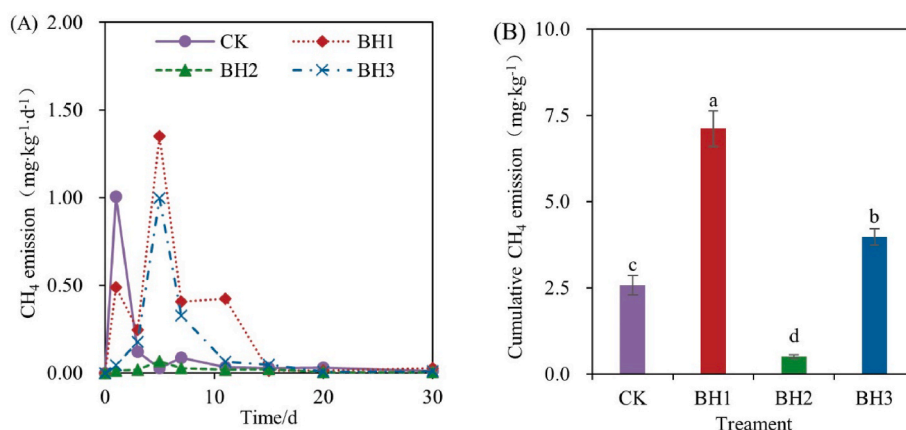


Fig. 4. The emission rate and cumulative emission of CH₄ during pig manure composting.

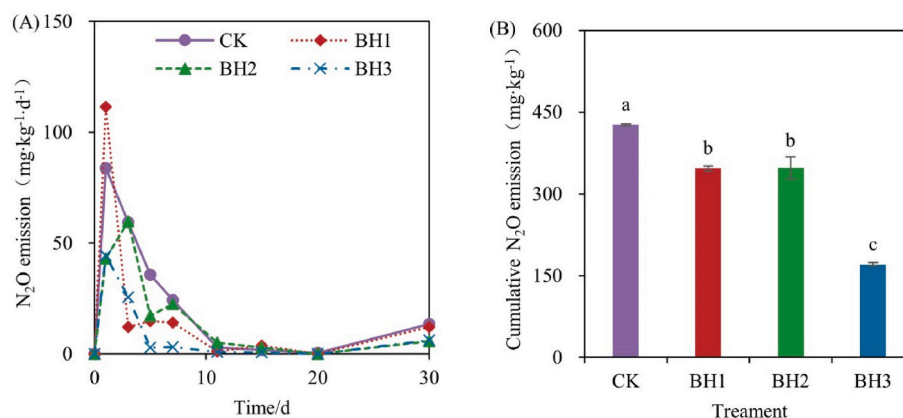


Fig. 5. The emission rate and cumulative emission of N_2O during pig manure composting.

results of other studies (Chen et al., 2018; Wang et al., 2022; Tang et al., 2023). Probably because of the decrease in NH_4^+-N content and the reduction in temperature, the activity of nitrifying bacteria increased, which promoted the production and emission of N_2O (Cáceres et al., 2018).

The cumulative N_2O emissions of the four treatments were further analyzed, and it was found that all three additions of Chinese herbal powder suppressed N_2O emissions (Fig. 5B). In the present investigation, the cumulative N_2O emissions ranged from 170.4 mg kg^{-1} to 426.6 mg kg^{-1} , which was in agreement with the cumulative emission reported in other studies (Yang et al., 2023). Compared with the control, the cumulative N_2O emissions in treatments BH1, BH2, and BH3 were reduced by 18.7%, 18.4%, and 60.0%, respectively, with significant differences ($p < 0.001$). The inhibition of N_2O emissions by the Chinese herbal powder may be attributed to (1) the optimization of the compost's pore structure, which inhibits denitrification (Agyarko-Mintah et al., 2017; Godlewska et al., 2017), and (2) the active substances in the herbs influencing the nitrogen cycle in the compost (Bardon et al., 2014). Therefore, the addition of Chinese herbs, particularly at 0.5% concentration, effectively inhibited N_2O emissions during the composting process, with the highest reduction observed in this treatment.

3.4. Effects of Chinese herbal plant powder on GWP of pig manure composting pile during whole composting

In the current study, the greenhouse effect due to GHG emissions during composting was further calculated in terms of CO_2 equivalent (Fig. 6). The CO_2 equivalent values for the treatments CK, BH1, BH2, and BH3 were 200.7, 196.4, 158.4, and $91.3 \text{ g kg}^{-1} CO_2\text{-eq}$, respectively; moreover, the greenhouse effects for treatments BH2 and BH3 were significantly lower than those of CK ($p < 0.05$). This indicates that the addition of herbs at proportions of 0.1% and 0.5% can reduce the greenhouse effect during livestock manure composting. The poor abatement effect of treatment BH1 is due to its promotion of CO_2 and CH_4 emissions. Treatment BH3, while promoting CH_4 emissions, suppresses CO_2 and N_2O emissions, and has a particularly good abatement effect on N_2O , which is a greenhouse gas with a high 100-year GWP (Shakoor et al., 2021; Y. Liu et al., 2023). This makes treatment BH3 the lowest in terms of GWP. This effect may be attributed to the presence of proanthocyanidins in the herbs which affect microbial activity related to carbon and nitrogen conversion in the composting process (Bardon et al., 2014, 2016; Wu et al., 2017; Huang et al., 2022).

4. Conclusions

The addition of 0.05% of the Chinese herbal powder promoted CO_2 and CH_4 emissions and suppressed N_2O emissions from compost. The addition of 0.1% herbal powder inhibited the emission of three

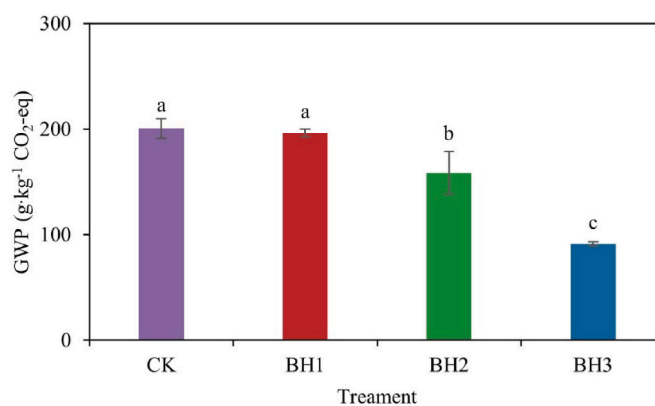


Fig. 6. Global warming potential during pig manure composting.

greenhouse gases simultaneously, although it also promoted CH_4 emissions. However, the addition of 0.1% demonstrated the best inhibitory effect on CO_2 and N_2O . The addition of both 0.1% and 0.5% herbal powder significantly reduced the global warming potential, with the 0.5% herbal powder showing the most substantial inhibitory effect.

CRediT authorship contribution statement

Ya-jun Duan: Writing – original draft, Visualization, Methodology, Conceptualization. **Hong-sheng Wu:** Writing – review & editing, Supervision, Funding acquisition. **Guang-yao Chen:** Investigation. **Feng-yi Hu:** Resources. **Yan-hui Li:** Visualization. **Zhen-wei Li:** Validation. **Jin-he Chi:** Conceptualization. **Dong-ming Lu:** Conceptualization. **Xiao-ming Gu:** Investigation. **Ming-chun Di:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The work was supported by Jiangsu Provincial Fund of Agricultural Resources and Ecological Protection Program from Chinese Central Government (SUNONGJI (2020)16 \ SUCAINONG (2020)33).

financially.

References

- Agyarko-Mintah, E., Cowie, Annette, Singh, B.P., Joseph, S., Van Zwieten, L., Cowie, Alan, Harden, S., Smillie, R., 2017. Biochar increases nitrogen retention and lowers greenhouse gas emissions when added to composting poultry litter. *Waste Manag.* 61, 138–149. <https://doi.org/10.1016/j.wasman.2016.11.027>.
- Awasthi, M.K., Duan, Y., Awasthi, S.K., Liu, T., Zhang, Z., 2020a. Influence of bamboo biochar on mitigating greenhouse gas emissions and nitrogen loss during poultry manure composting. *Bioresour. Technol.* 303, 122952 <https://doi.org/10.1016/j.biortech.2020.122952>.
- Awasthi, M.K., Duan, Y., Awasthi, S.K., Liu, T., Zhang, Z., Kim, S.-H., Pandey, A., 2020b. Effect of biochar on emission, maturity and bacterial dynamics during sheep manure composting. *Renew. Energy* 152, 421–429. <https://doi.org/10.1016/j.renene.2020.01.065>.
- Ba, S., Qu, Q., Zhang, K., Groot, J.C.J., 2020. Meta-analysis of greenhouse gas and ammonia emissions from dairy manure composting. *Biosyst. Eng.* 193, 126–137. <https://doi.org/10.1016/j.biosystemseng.2020.02.015>.
- Bai, M., Impraim, R., Coates, T., Flesch, T., Trouvé, R., Van Grinsven, H., Cao, Y., Hill, J., Chen, D., 2020. Lignite effects on NH₃, N₂O, CO₂ and CH₄ emissions during composting of manure. *J. Environ. Manage.* 271, 110960 <https://doi.org/10.1016/j.jenvman.2020.110960>.
- Bardon, C., Piola, F., Bellvert, F., Haichar, F. el Z., Comte, G., Meiffren, G., Pommier, T., Puijalon, S., Tsafack, N., Poly, F., 2014. Evidence for biological denitrification inhibition (BDI) by plant secondary metabolites. *New Phytol.* 204, 620–630. <https://doi.org/10.1111/nph.12944>.
- Bardon, C., Poly, F., Piola, F., Pancton, M., Comte, G., Meiffren, G., Haichar, F. el Z., 2016. Mechanism of biological denitrification inhibition: procyanidins induce an allosteric transition of the membrane-bound nitrate reductase through membrane alteration. *FEMS Microbiol. Ecol.* 92, fiw034 <https://doi.org/10.1093/femsec/fiw034>.
- Cáceres, R., Malińska, K., Marfà, O., 2018. Nitrification within composting: a review. *Waste Manag.* 72, 119–137. <https://doi.org/10.1016/j.wasman.2017.10.049>.
- Cao, Y., Wang, X., Bai, Z., Chadwick, D., Misselbrook, T.G., Sommer, S., Qin, W., Ma, L., 2019. Mitigation of ammonia, nitrous oxide and methane emissions during solid waste composting with different additives: a meta-analysis. *J. Clean. Prod.* 235, 626–635. <https://doi.org/10.1016/j.jclepro.2019.06.288>.
- Chen, H., Awasthi, M.K., Liu, T., Zhao, J., Ren, X., Wang, M., Duan, Y., Awasthi, S.K., Zhang, Z., 2018. Influence of clay as additive on greenhouse gases emission and maturity evaluation during chicken manure composting. *Bioresour. Technol.* 266, 82–88. <https://doi.org/10.1016/j.biortech.2018.06.073>.
- Chen, H., Awasthi, S.K., Liu, T., Duan, Y., Ren, X., Zhang, Z., Pandey, A., Awasthi, M.K., 2020. Effects of microbial culture and chicken manure biochar on compost maturity and greenhouse gas emissions during chicken manure composting. *J. Hazard Mater.* 389, 121908 <https://doi.org/10.1016/j.jhazmat.2019.121908>.
- Chen, P., Zheng, X., Cheng, W., 2022. Biochar combined with ferrous sulfate reduces nitrogen and carbon losses during agricultural waste composting and enhances microbial diversity. *Process Saf. Environ. Prot.* 162, 531–542. <https://doi.org/10.1016/j.psep.2022.04.042>.
- Czekala, W., Malińska, K., Cáceres, R., Janczak, D., Dach, J., Lewicki, A., 2016. Co-composting of poultry manure mixtures amended with biochar – the effect of biochar on temperature and C-CO₂ emission. *Bioresour. Technol.* 200, 921–927. <https://doi.org/10.1016/j.biortech.2015.11.019>.
- Fan, X., Chen, X., Chen, T., Liu, X., Song, Y., Tan, S., Chen, Y., Yan, P., Wang, X., 2023. Effects of substituting synthetic nitrogen with organic amendments on crop yield, net greenhouse gas emissions and carbon footprint: a global meta-analysis. *Field Crops Res.* 301, 109035 <https://doi.org/10.1016/j.fcr.2023.109035>.
- Ge, J., Huang, G., Li, J., Han, L., 2018. Particle-scale visualization of the evolution of methanogens and methanotrophs and its correlation with CH₄ emissions during manure aerobic composting. *Waste Manag.* 78, 135–143. <https://doi.org/10.1016/j.wasman.2018.05.045>.
- Godlewska, P., Schmidt, H.P., Ok, Y.S., Oleszczuk, P., 2017. Biochar for composting improvement and contaminants reduction. A review. *Bioresour. Technol.* 246, 193–202. <https://doi.org/10.1016/j.biortech.2017.07.095>.
- Harindintwali, J.D., Zhou, J., Muhoza, B., Wang, F., Herzberger, A., Yu, X., 2021. Integrated eco-strategies towards sustainable carbon and nitrogen cycling in agriculture. *J. Environ. Manage.* 293, 112856 <https://doi.org/10.1016/j.jenvman.2021.112856>.
- Harrison, B.P., Gao, S., Gonzales, M., Thao, T., Bischak, E., Ghezzehei, T.A., Berhe, A.A., Diaz, G., Ryals, R.A., 2022. Dairy manure Co-composting with wood biochar plays a critical role in meeting global methane goals. *Environ. Sci. Technol.* 56, 10987–10996. <https://doi.org/10.1021/acs.est.2c03467>.
- Hoang, H.G., Thuy, B.T.P., Lin, C., Vo, D.-V.N., Tran, H.T., Bahari, M.B., Le, V.G., Vu, C. T., 2022. The nitrogen cycle and mitigation strategies for nitrogen loss during organic waste composting: a review. *Chemosphere* 300, 134514. <https://doi.org/10.1016/j.chemosphere.2022.134514>.
- Huang, D., Gao, L., Cheng, M., Yan, M., Zhang, G., Chen, S., Du, L., Wang, G., Li, R., Tao, J., Zhou, W., Yin, L., 2022. Carbon and N conservation during composting: a review. *Sci. Total Environ.* 840, 156355 <https://doi.org/10.1016/j.scitotenv.2022.156355>.
- Jiang, J., Kang, K., Wang, C., Sun, X., Dang, S., Wang, N., Wang, Y., Zhang, C., Yan, G., Li, Y., 2018. Evaluation of total greenhouse gas emissions during sewage sludge composting by the different dicyandiamide added forms: mixing, surface broadcasting, and their combination. *Waste Manag.* 81, 94–103. <https://doi.org/10.1016/j.wasman.2018.10.003>.
- Kraut-Cohen, J., Zolti, A., Rotbart, N., Bar-Tal, A., Laor, Y., Medina, S., Shawahna, R., Saadi, I., Raviv, M., Green, S.J., Yermiyahu, U., Minz, D., 2023. Short- and long-term effects of continuous compost amendment on soil microbiome community. *Comput. Struct. Biotechnol. J.* 21, 3280–3292. <https://doi.org/10.1016/j.csbj.2023.05.030>.
- Kuypers, M.M.M., Marchant, H.K., Kartal, B., 2018. The microbial nitrogen-cycling network. *Nat. Rev. Microbiol.* 16, 263–276. <https://doi.org/10.1038/nrmicro.2018.9>.
- Lei, L., Gu, J., Wang, X., Song, Z., Yu, J., Guo, H., Xie, J., Wang, J., Sun, W., 2022. Effects and microbial mechanisms of phosphogypsum and medical stone on organic matter degradation and methane emissions during swine manure composting. *J. Environ. Manage.* 315, 115139 <https://doi.org/10.1016/j.jenvman.2022.115139>.
- Li, G., Niu, W., Ma, L., Du, Y., Zhang, Q., Sun, J., Siddique, K.H.M., 2023. Legacy effects of wheat season organic fertilizer addition on microbial co-occurrence networks, soil function, and yield of the subsequent maize season in a wheat-maize rotation system. *J. Environ. Manage.* 347, 119160 <https://doi.org/10.1016/j.jenvman.2023.119160>.
- Liu, Q., He, X., Wang, K., Li, D., 2023. Biochar drives humus formation during composting by regulating the specialized metabolic features of microbiome. *Chem. Eng. J.* 141380 <https://doi.org/10.1016/j.cej.2023.141380>.
- Liu, T., Kumar Awasthi, M., Verma, S., Qin, S., Awasthi, S.K., Liu, H., Zhou, Y., Zhang, Z., 2021. Evaluation of cornstark as bulking agent on greenhouse gases emission and bacterial community during further composting. *Bioresour. Technol.* 340, 125713 <https://doi.org/10.1016/j.biortech.2021.125713>.
- Liu, Y., Ma, R., Li, D., Qi, C., Han, L., Chen, M., Fu, F., Yuan, J., Li, G., 2020. Effects of calcium magnesium phosphate fertilizer, biochar and spent mushroom substrate on compost maturity and gaseous emissions during pig manure composting. *J. Environ. Manage.* 267, 110649 <https://doi.org/10.1016/j.jenvman.2020.110649>.
- Liu, Y., Tang, R., Li, L., Zheng, G., Wang, J., Wang, G., Bao, Z., Yin, Z., Li, G., Yuan, J., 2023. A global meta-analysis of greenhouse gas emissions and carbon and nitrogen losses during livestock manure composting: influencing factors and mitigation strategies. *Sci. Total Environ.* 163900 <https://doi.org/10.1016/j.scitotenv.2023.163900>.
- Liu, Y., Tang, R., Liu, G., Li, G., Wang, J., Cai, Y., Kong, Y., Yuan, J., 2024. Dicyandiamide addition delay nitrous oxide emission and shift its production pathway from denitrification to incomplete nitrification in maturation phase of composting. *Chem. Eng. J.* 495, 153225 <https://doi.org/10.1016/j.cej.2024.153225>.
- Luo, Y., Liang, J., Zeng, G., Chen, M., Mo, D., Li, G., Zhang, D., 2018. Seed germination test for toxicity evaluation of compost: its roles, problems and prospects. *Waste Manag.* 71, 109–114. <https://doi.org/10.1016/j.wasman.2017.09.023>.
- National Bureau of Statistics of China, 2022. *China Rural Statistical Yearbook*. China Statistics Press, Beijing.
- Nguyen, M.K., Lin, C., Hoang, H.G., Sanderson, P., Dang, B.T., Bui, X.T., Nguyen, N.S.H., Vo, D.-V.N., Tran, H.T., 2022. Evaluate the role of biochar during the organic waste composting process: a critical review. *Chemosphere* 299, 134488. <https://doi.org/10.1016/j.chemosphere.2022.134488>.
- Niu, Y., Guo, S., Han, Y., Si, Y., Li, P., Li, F., 2024. Fulvic acid and fermentation agent optimize in situ composting by reducing antibiotic resistance genes abundances and altering succession of bacterial communities. *Int. Biodeterior. Biodegrad.* 191, 105813 <https://doi.org/10.1016/j.ibiod.2024.105813>.
- Ottani, F., Parenti, M., Santunione, G., Moscatelli, G., Kahn, R., Pedrazzi, S., Allesina, G., 2023. Effects of different gasification biochar grain size on greenhouse gases and ammonia emissions in municipal aerated composting processes. *J. Environ. Manage.* 331, 117257 <https://doi.org/10.1016/j.jenvman.2023.117257>.
- Qiu, Z., Li, M., Song, L., Wang, C., Yang, S., Yan, Z., Wang, Y., 2021. Study on nitrogen-retaining microbial agent to reduce nitrogen loss during chicken manure composting and nitrogen transformation mechanism. *J. Clean. Prod.* 285, 124813 <https://doi.org/10.1016/j.jclepro.2020.124813>.
- Ren, X., Jiao, M., Zhang, Z., Syed, A., Bahkali, A.H., 2023. The efficient solution to decline the greenhouses emission and enrich the bacterial community during pig manure composting: regulating the particle size of cornstark. *Bioresour. Technol.* 387, 129596 <https://doi.org/10.1016/j.biortech.2023.129596>.
- Shakoor, A., Shakoor, S., Rehman, A., Ashraf, F., Abdullah, M., Shahzad, S.M., Farooq, T. H., Ashraf, M., Manzoor, M.A., Altaf, M.M., Altaf, M.A., 2021. Effect of animal manure, crop type, climate zone, and soil attributes on greenhouse gas emissions from agricultural soils—a global meta-analysis. *J. Clean. Prod.* 278, 124019 <https://doi.org/10.1016/j.jclepro.2020.124019>.
- Tang, R., Liu, Y., Ma, R., Zhang, L., Li, Y., Li, G., Wang, Dingmei, Lin, J., Li, Q., Yuan, J., 2023. Effect of moisture content, aeration rate, and C/N on maturity and gaseous emissions during kitchen waste rapid composting. *J. Environ. Manage.* 326, 116662 <https://doi.org/10.1016/j.jenvman.2022.116662>.
- Van den Bergh, S.G., Chardon, I., Meima-Franke, M., Costa, O.Y.A., Korthals, G.W., de Boer, W., Bodellier, P.L.E., 2023. The intrinsic methane mitigation potential and associated microbes add product value to compost. *Waste Manag.* 170, 17–32. <https://doi.org/10.1016/j.wasman.2023.07.027>.
- Wang, N., Awasthi, M.K., Pan, J., Jiang, S., Wan, F., Lin, X., Yan, B., Zhang, J., Zhang, L., Huang, H., Li, H., 2022. Effects of biochar and biogas residue amendments on N₂O emission, enzyme activities and functional genes related with nitrification and denitrification during rice straw composting. *Bioresour. Technol.* 357, 127359 <https://doi.org/10.1016/j.biortech.2022.127359>.
- Wang, N., He, Y., Zhao, K., Lin, X., He, X., Chen, A., Wu, G., Zhang, J., Yan, B., Luo, L., Xu, D., 2024. Greenhouse gas emission characteristics and influencing factors of agricultural waste composting process: a review. *J. Environ. Manage.* 354, 120337 <https://doi.org/10.1016/j.jenvman.2024.120337>.

- Wu, H.-S., Chen, S.-Y., Li, J., Liu, D.-Y., Zhou, J., Xu, Y., Shang, X.-X., Wei, D., Yu, L., Fang, X., Li, S., Wang, K., 2017. An approach to mitigating soil CO₂ emission by biochemically inhibiting cellulolytic microbial populations through mediation via the medicinal herb *Isatis indigotica*. *Atmos. Environ.* 158, 259–269. <https://doi.org/10.1016/j.atmosenv.2017.03.046>.
- Yang, Y., Chen, W., Liu, G., Kong, Y., Wang, G., Yin, Z., Li, G., Yuan, J., 2023. Effects of cornstalk and sawdust coverings on greenhouse gas emissions during sheep manure storage. *Waste Manag.* 166, 104–114. <https://doi.org/10.1016/j.wasman.2023.04.034>.
- Yang, Y., Yin, Z., Li, L., Li, Y., Liu, Y., Luo, Y., Li, G., Yuan, J., 2022. Effects of dicyandiamide, phosphogypsum and superphosphate on greenhouse gas emissions during pig manure composting. *Sci. Total Environ.* 846, 157487 <https://doi.org/10.1016/j.scitotenv.2022.157487>.
- Yin, Y., Gu, M., Zhang, W., Yang, C., Li, H., Wang, X., Chen, R., 2024. Relationships between different types of biochar and N₂O emissions during composting based on roles of nosZ-carrying denitrifying bacterial communities enriched on compost and biochar particles. *Bioresour. Technol.* 394, 130214 <https://doi.org/10.1016/j.biortech.2023.130214>.
- Yu, J., Gu, J., Wang, X., Lei, L., Guo, H., Song, Z., Sun, W., 2023. Exploring the mechanism associated with methane emissions during composting: inoculation with lignocellulose-degrading microorganisms. *J. Environ. Manage.* 325, 116421 <https://doi.org/10.1016/j.jenvman.2022.116421>.
- Zhang, Z., Liu, D., Qiao, Y., Li, S., Chen, Y., Hu, C., 2021. Mitigation of carbon and nitrogen losses during pig manure composting: a meta-analysis. *Sci. Total Environ.* 783, 147103 <https://doi.org/10.1016/j.scitotenv.2021.147103>.
- Zhou, Y., Selvam, A., Wong, J.W.C., 2018. Chinese medicinal herbal residues as a bulking agent for food waste composting. *Bioresour. Technol.* 249, 182–188. <https://doi.org/10.1016/j.biortech.2017.09.212>.
- Zhu, Z., Wu, S., Qi, B., Luo, J., Wan, Y., 2022. Highly efficient glucose production from raw non-pretreated Chinese medicinal herbal residues via the synergism of cellulase and amylolytic enzymes. *Bioresour. Technol.* 364, 128102 <https://doi.org/10.1016/j.biortech.2022.128102>.